

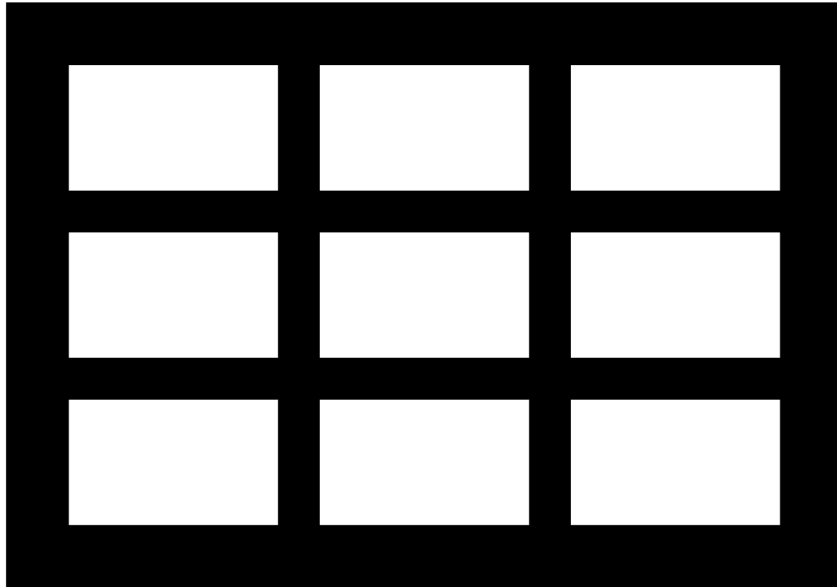
4COSC002W Mathematics for Computing

Lecture 7

Introduction to Matrices. Types of Matrices. Operations with
Matrices

UNIVERSITY OF
WESTMINSTER

What is a Matrix?



A *matrix* is a rectangular array of numbers, symbols, or expressions arranged in rows and columns.

The individual items in a matrix are called its *elements* or entries.

Matrix Notation

A matrix is usually denoted by a **capital letter** (A, B, C, ...) and its elements by the corresponding **lowercase letter with two subscript indices** (a_{mn}), where:

- The first subscript (m) denotes the row number.
- The second (n) subscript denotes the column number.

For example:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Here, **A** is an **m**×**n** matrix with **m** rows and **n** columns.

Matrix Order

The *order* of a matrix is the number of rows and columns it has.

For Example:

$$M = \begin{bmatrix} 4 & -1 \\ 0 & 3 \\ 7 & 5 \end{bmatrix}$$

A 3×2 matrix is a matrix that has 3 rows and 2 columns

In this matrix, M :

- The element in the first row, first column is 4 (M_{11}).
- The element in the first row, second column is -1 (M_{12}).
- The element in the second row, first column is 0 (M_{21}).
- The element in the second row, second column is 3 (M_{22}).
- The element in the third row, first column is 7 (M_{31}).
- The element in the third row, second column is 5 (M_{32}).

Types of Matrices

- **Row Matrix:** A matrix with only one row.
- **Column Matrix:** A matrix with only one column.
- **Square Matrix:** A matrix with the same number of rows and columns.
- **Zero Matrix:** A matrix with all elements equal to zero.
- **Identity Matrix:** A square matrix with 1's on the diagonal and 0's elsewhere.
- **Diagonal Matrix:** A square matrix where all elements off the main diagonal are zero.
- **Triangular Matrix:** A square matrix with zero elements above (upper triangular) or below (lower triangular) the main diagonal.

Square Matrix

A *square matrix* has the same number of rows and columns ($n \times n$). Matrices for which $m \neq n$ are called *non-square*.

Example:

$$S = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

This is a 2×2 square matrix.

Square matrices have special properties (some explained in next week's lecture notes) and are particularly important in linear algebra.

Many operations, such as taking *determinants* and computing *eigenvalues*, are only defined for square matrices.

Identity Matrix (Unit Matrix)

An *identity matrix* is a square matrix where all elements on the main diagonal are 1, and all off-diagonal elements are 0.

Example:

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This is a 2×2 identity matrix.

The subscript denotes its size.

Identity Matrix (Unit Matrix)

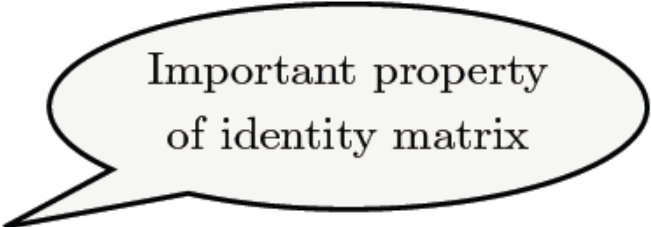
It is the multiplicative identity in matrix algebra, meaning any matrix multiplied by the identity matrix will result in the original matrix.

If A is an $m \times n$ matrix, then

$$I_m A = A \quad \text{and} \quad A I_n = A.$$

If A is a square matrix, then

$$IA = A = AI.$$



Important property
of identity matrix

Diagonal Matrix

A *diagonal matrix* is a square matrix where all elements off the main diagonal are zero, but the diagonal can contain non-zero elements.

Example:

$$D = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$$

This is a 3×3 diagonal matrix.

Triangular Matrix

A *triangular matrix* is a type of square matrix where all elements above the main diagonal are zero in a lower triangular matrix or all elements below the main diagonal are zero in an upper triangular matrix.

Lower Triangular Matrix Example:

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

Upper Triangular Matrix Example:

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Row Matrix (Row Vector)

A *row matrix* has only one row and a certain number of columns (n). It is also referred to as a *row vector*.

Example:

$$R = [1 \quad -7 \quad 13]$$

This is a 1×3 row matrix.

Column Matrix (Column Vector)

Conversely, a *column matrix* has only one column and a certain number of rows (m). It is also called a *column vector*.

Example:

$$C = \begin{bmatrix} 7 \\ -1 \\ 4 \end{bmatrix}$$

This is a 3×1 column matrix.

Zero Matrix (Null Matrix)

A *zero matrix* is a matrix where all elements are **zero**. It can be of any dimension $m \times n$.

Example:

$$Z = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

This is a 2×2 zero matrix,
but zero matrices can be of any size.

ACTIVITY: Identify the Types of Matrices

Matrix A

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Matrix B

$$\begin{bmatrix} 2 & 4 \\ -1 & 3 \\ 5 & 0 \end{bmatrix}$$

Matrix C

$$\begin{bmatrix} 0 & 4 \\ 7 & 8 \end{bmatrix}$$

Matrix D

$$\begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

Matrix E

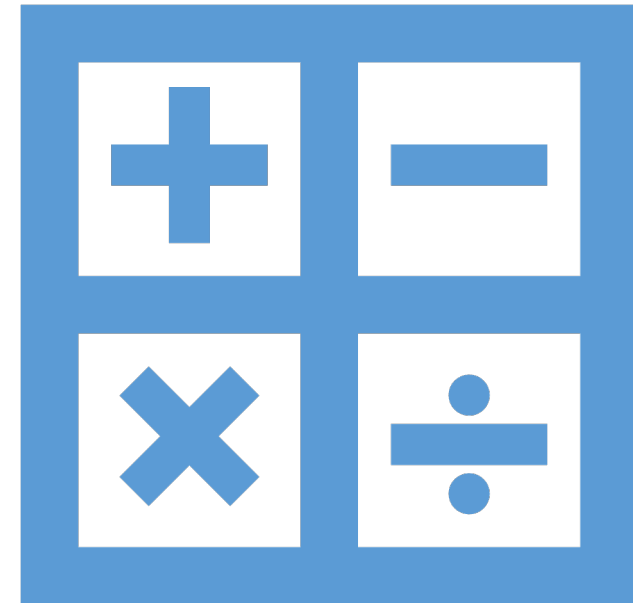
$$\begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix}$$

Operations with Matrices

Similarly to addition, subtraction, multiplication, etc. on numbers, we can also perform these operations (and others) on matrices.

The operations we will now look at are:

- ✓ addition, subtraction
- ✓ multiplication by scalar
- ✓ matrix transpose
- ✓ multiplication of matrices



Addition / Subtraction

To add matrices, we add numbers in corresponding positions. However, to enable addition, matrices must be of the same order.

For example, you cannot add a 3x3 matrix and a 4x4 matrix because their numbers of rows and columns do not match

The addition of matrices **A** and **B** is written as **A + B**. If matrix **C** results from the addition of matrices **A** and **B**, then we write **C = A + B**.

If C is the sum of two m x n matrices, A and B, then C is an m x n matrix with elements:

$$c_{ij} = a_{ij} + b_{ij}, \text{ where } i = 1, 2, \dots, m \text{ and } j = 1, 2, \dots, n$$

Example: Addition

Given matrices A and B :

$$A = \begin{bmatrix} 5 & 7 \\ 3 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$$

Addition $A + B$:

$$A + B = \begin{bmatrix} 5 + 2 & 7 + 4 \\ 3 + 1 & 2 + 3 \end{bmatrix} = \begin{bmatrix} 7 & 11 \\ 4 & 5 \end{bmatrix}$$

Subtraction $A - B$:

$$A - B = \begin{bmatrix} 5 - 2 & 7 - 4 \\ 3 - 1 & 2 - 3 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 2 & -1 \end{bmatrix}$$

Example: Aggregating Sensor Data Using Matrix Addition and Subtraction

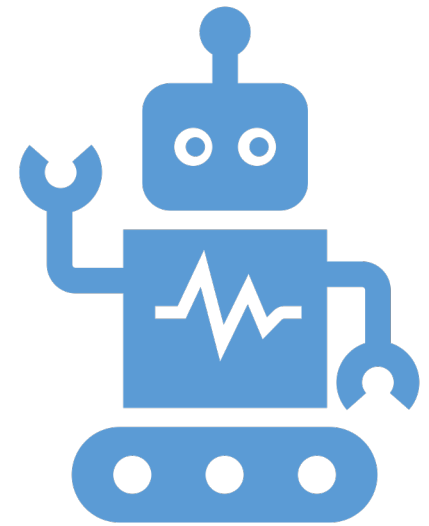
Imagine you have a robot equipped with multiple sensors that collect environmental data (like temperature and humidity) from different locations. Each matrix represents readings from a different time point, and each element is the data collected from a specific location.

Matrix A: Sensor Data at Time 1

$$A = \begin{bmatrix} 22 & 24 \\ 20 & 25 \end{bmatrix}$$

Matrix B: Sensor Data at Time 2

$$B = \begin{bmatrix} 23 & 25 \\ 21 & 26 \end{bmatrix}$$



Example: Aggregating Sensor Data Using Matrix Addition and Subtraction

1. Combining Sensor Data (Addition)

To get the *total data* collected over both time points, you can add the matrices element-wise:

$$A + B = \begin{bmatrix} 22 + 23 & 24 + 25 \\ 20 + 21 & 25 + 26 \end{bmatrix} = \begin{bmatrix} 45 & 49 \\ 41 & 51 \end{bmatrix}$$

The purpose of adding the data is to get an aggregate reading from each sensor, which helps track total resource usage or overall environmental impact over time. This cumulative data provides insights into conditions like temperature or humidity across different periods.

2. Calculating the Change in Data (Subtraction)

To find the *change* in sensor readings between the two time points (e.g., to detect environmental shifts), you can subtract A from B :

$$B - A = \begin{bmatrix} 23 - 22 & 25 - 24 \\ 21 - 20 & 26 - 25 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

The resulting matrix shows the increase in readings for each sensor location between the two times.

Multiplication by a Scalar

We can multiply a matrix **by a scalar** (scalar means number, constant)

This operation means that *each element* of the matrix is multiplied by the scalar.

Example:

Given *matrix A*: $A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$

Multiply by *scalar 3*: $3 \times A = \begin{bmatrix} 3 \times 2 & 3 \times 4 \\ 3 \times 6 & 3 \times 8 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 18 & 24 \end{bmatrix}$

Example: Adjusting Image Brightness Using Scalar Multiplication

In digital images, brightness is controlled by the pixel intensity values, where each pixel's value represents its brightness. For a grayscale image, these values typically range from 0 (black) to 255 (white).

Imagine you have a small grayscale image represented by a matrix, where each value in the matrix is the brightness of a pixel.

Original Image Matrix I:

$$I = \begin{bmatrix} 50 & 100 & 150 \\ 80 & 120 & 160 \\ 90 & 140 & 200 \end{bmatrix}$$

Each value here represents the brightness of a pixel, with 50 being quite dark, 200 being quite bright, and values in between representing varying shades of gray.

Goal: Increase the Brightness by 20%



Example: Adjusting Image Brightness Using Scalar Multiplication

To increase the brightness of the entire image by 20%, we can use scalar multiplication.

In this case, our scalar factor will be 1.2, meaning each pixel value will be multiplied by 1.2, or 120% of its original value.

Solution: We will multiply each element in the *matrix I* by the scalar 1.2.

$$I_{\text{brightened}} = 1.2 \times \begin{bmatrix} 50 & 100 & 150 \\ 80 & 120 & 160 \\ 90 & 140 & 200 \end{bmatrix} = \begin{bmatrix} 60 & 120 & 180 \\ 96 & 144 & 192 \\ 108 & 168 & 240 \end{bmatrix}$$

Matrix Transpose

The *transpose of a matrix* is a new matrix that simply has the rows and columns “exchanged”.

We denote the transpose of matrix A as A^T . If A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix.

Example:

Given matrix A : $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

To find the transpose of A , denoted as A^T , swap the rows and columns: $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

Example: Student Grades in Different Subjects

Imagine you have a matrix that represents the grades of students in three different subjects. Each row represents a student, and each column represents a subject.

Matrix G: Original Grades Matrix

$$G = \begin{bmatrix} 85 & 90 & 78 \\ 88 & 76 & 92 \\ 90 & 85 & 80 \end{bmatrix}$$

Rows:

- Row 1: Grades of Student 1 in three subjects.
- Row 2: Grades of Student 2 in three subjects.
- Row 3: Grades of Student 3 in three subjects.

Goal: Transpose the matrix to organise data by subjects



Example:

Student Grades in Different Subjects

By transposing, we can reorganize the matrix so that each row now represents a subject, and each column represents the grades of all students in that subject.

Transpose of G :

$$G = \begin{bmatrix} 85 & 90 & 78 \\ 88 & 76 & 92 \\ 90 & 85 & 80 \end{bmatrix}; G^T = \begin{bmatrix} 85 & 88 & 90 \\ 90 & 76 & 85 \\ 78 & 92 & 80 \end{bmatrix}$$

Rows (After Transpose):

- Row 1: Grades in Subject 1 for all students.
- Row 2: Grades in Subject 2 for all students.
- Row 3: Grades in Subject 3 for all students.

Symmetric Matrix

A *symmetric matrix* is a square matrix that is equal to its transpose.

This means that the element in the i^{th} row and j^{th} column equals the element in the j^{th} row and i^{th} column.

Example:

$$Sym = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$$

For the *matrix Sym* to be symmetric,
it must hold that $Sym = Sym^T$.

Matrix Equality

Two matrices are considered *equal* if they have the same dimensions and their corresponding elements are equal.

For matrix equality to hold, every entry in one matrix must match the corresponding entry in the other matrix.

Example:

Matrix A:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Matrix B:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Matrix C:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

A=B, but A≠C (same elements, but dimensions are not the same)

ACTIVITY: Matrix Equality

Given that the following matrices are equal, find the values of x , y , and z .

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ \frac{2}{3} & -5 \\ 6 & 4 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} x & 1 \\ \frac{2}{3} & y-10 \\ \frac{z}{2} & 4 \end{bmatrix}$$

ACTIVITY: Answer

Since $A = B$, all their elements are equal.

Compare the (1,1) elements: $x = 2$

Compare the (2,2) elements: $y - 10 = -5$

Solving for y : $y = -5 + 10 = 5$

Compare the (3,1) elements: $\frac{z}{2} = 6$

Solving for z : $z = 6 \times 2 = 12$

Final Values: $x = 2, y = 5, z = 12$

Matrix Multiplication

Matrix multiplication is a binary operation that takes a pair of matrices and produces another matrix. This operation is not element-wise like addition or subtraction but rather involves a sequence of products and sums.

If we have two matrices **A** and **B** then we represent their product as **AB**, or **A×B**.

Two matrices can only be multiplied if their dimensions match in a specific way: we can multiply two matrices if the number of columns in the first matrix matches the number of rows in the second matrix.

Matrix multiplication is NOT commutative! That is, usually **AB ≠ BA**

Matrix Multiplication

If we have two matrices – one of order $m \times n$ and the other of order $p \times q$ then we can only multiply them if $n = p$



$m \times q$ will be the size of the resulting matrix – sometimes called the ‘outer dimensions’

Example: Matrix Multiplication

1. Let **A** be 2×2 and matrix **B** be 2×3 matrices. What is their product?

Answer: They can be multiplied, and the product will be matrix **AB** of the order 2×3 .

2. Let **C** be 3×3 , and **D** be 4×3 . What is their product?

Answer: We cannot do this!

Matrix Multiplication

The steps for multiplying two matrices, A and B :

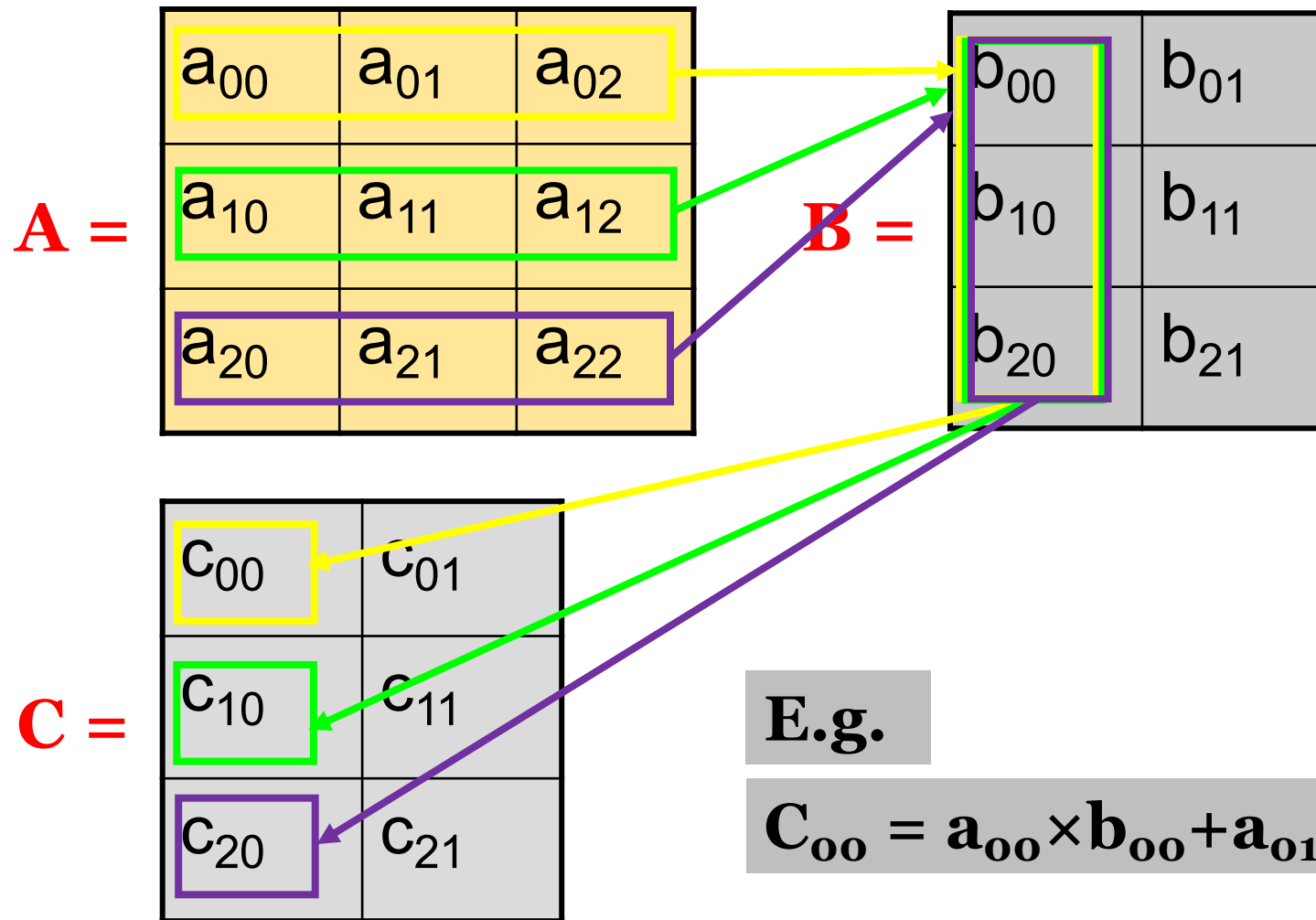
- The number of columns in the first matrix must equal the number of rows in the second matrix.
- The element at the position (i, j) in the resulting matrix C is calculated by taking the dot product of the i -th row of A and the j -th column of B .
- The dot product mentioned in step 2 is calculated by multiplying corresponding elements from the i -th row of A and the j -th column of B and then summing those products.

The resulting matrix has its elements defined as:

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

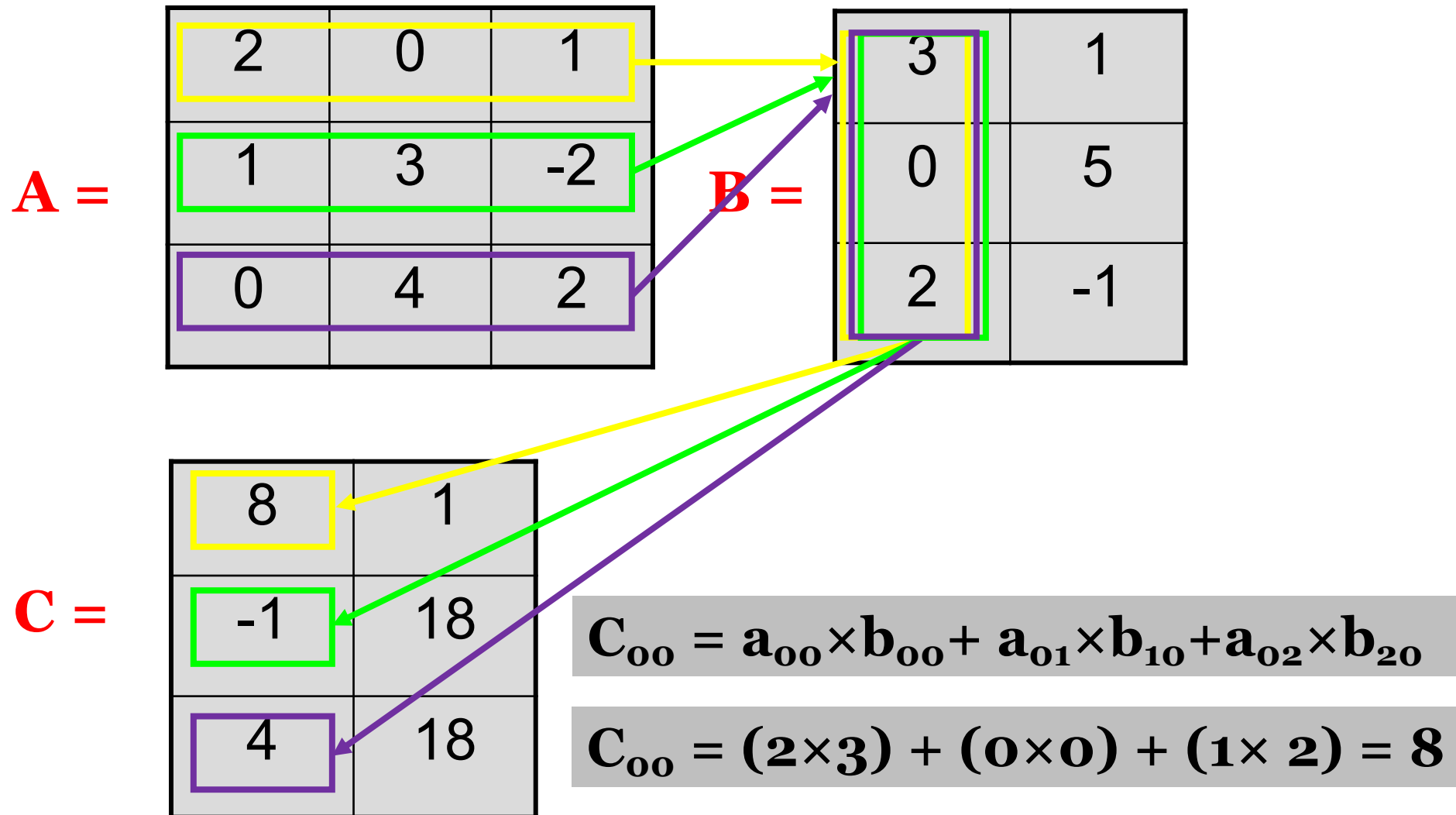
Where c_{ij} is the element in the i -th row and j -th column of matrix C , a_{ik} is the element in the i -th row and k -th column of matrix A , and b_{kj} is the element in the k -th row and j -th column of matrix B .

Multiplication Process: $C = AB$



NB: Indexing in this example starts with 00

Example: Multiplication Process



Example: Multiplication Process

The resulting **matrix C** will be a 3×2 matrix since A is 3×3 and B is 3×2 .

We calculate each element C_{ij} by taking the dot product of the i -th row of A with the j -th column of B:

$$C_{00} = (2 \times 3) + (0 \times 0) + (1 \times 2) = 6 + 0 + 2 = 8$$

$$C_{01} = (2 \times 1) + (0 \times 5) + (1 \times -1) = 2 + 0 - 1 = 1$$

$$C_{10} = (1 \times 3) + (3 \times 0) + (-2 \times 2) = 3 + 0 - 4 = -1$$

$$C_{11} = (1 \times 1) + (3 \times 5) + (-2 \times -1) = 1 + 15 + 2 = 18$$

$$C_{20} = (0 \times 3) + (4 \times 0) + (2 \times 2) = 0 + 0 + 4 = 4$$

$$C_{21} = (0 \times 1) + (4 \times 5) + (2 \times -1) = 0 + 20 - 2 = 18$$

Example: Moving Characters Down on a Grid Using Matrix Multiplication

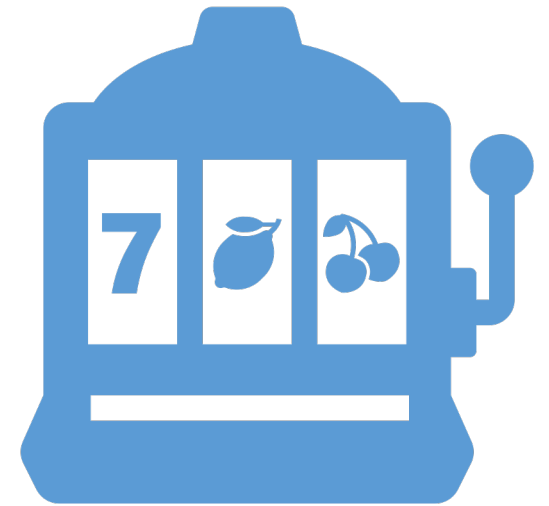
In this task, you'll learn how to move positions in a grid using *matrix multiplication*, a concept used in computer graphics and simple game design.

Imagine we have a *3x3 grid* where each *position* can either contain a *character* (represented by "1") or be *empty* (represented by "0").

Initial Position Matrix:

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Goal: Shift All Characters One Row Down



Example: Moving Characters Down on a Grid Using Matrix Multiplication

To move every character down by one row, we'll use a **shift matrix S**. This matrix will “push” each row down one position, wrapping the last row back to the top.

Shift Matrix S:

$$S = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Here's how each row of **S** accomplishes this:

- The first row of S ($[0, 0, 1]$) “pulls” the third row of P into the first row position of the result, wrapping it around.
- The second row of S ($[1, 0, 0]$) moves the first row of P into the second row.
- The third row of S ($[0, 1, 0]$) shifts the second row of P down to the third row.

Example: Moving Characters Down on a Grid Using Matrix Multiplication

We will multiply **P** by **S** (i.e., $P \times S$) to produce a new matrix **P_{shifted}** that represents the positions after shifting all characters down

First Row of $P_{shifted}$:

- Position (1,1): $(1 \times 0) + (1 \times 1) + (0 \times 0) = 0 + 1 + 0 = 1$
- Position (1,2): $(1 \times 0) + (1 \times 0) + (0 \times 1) = 0 + 0 + 0 = 0$
- Position (1,3): $(1 \times 1) + (1 \times 0) + (0 \times 0) = 1 + 0 + 0 = 1$

Second Row of $P_{shifted}$:

- Position (2,1): $(0 \times 0) + (1 \times 1) + (1 \times 0) = 0 + 1 + 0 = 1$
- Position (2,2): $(0 \times 0) + (1 \times 0) + (1 \times 1) = 0 + 0 + 1 = 1$
- Position (2,3): $(0 \times 1) + (1 \times 0) + (1 \times 0) = 0 + 0 + 0 = 0$

Third Row of $P_{shifted}$:

- Position (3,1): $(1 \times 0) + (0 \times 1) + (1 \times 0) = 0 + 0 + 0 = 0$
- Position (3,2): $(1 \times 0) + (0 \times 0) + (1 \times 1) = 0 + 0 + 1 = 1$
- Position (3,3): $(1 \times 1) + (0 \times 0) + (1 \times 0) = 1 + 0 + 0 = 1$

After performing all these calculations, we get the final *shifted matrix*:

$$P \times S = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P_{shifted} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$